

Frequency Analysis of Monoalphabetic Ciphers

**INSERT
VIDEO
HERE**

Frequency Analysis of Monoalphabetic Ciphers

**INSERT
VIDEO
HERE**

- Lesson 1 – Single-character frequencies.
 - Overview

Frequency Analysis of Monoalphabetic Ciphers

**INSERT
VIDEO
HERE**

- Lesson 1 – Single-character frequencies.
 - Overview
 - Practical hurdles
- Lesson 2 – Multi-character frequencies.
 - Digraphs
 - Repeated digraphs
- Lesson 3 – Example
 - Short ciphertext
 - Longer ciphertext

Frequency Analysis of Monoalphabetic Ciphers

**INSERT
VIDEO
HERE**

- Lesson 1 – Single-character frequencies.
 - Overview
 - Practical hurdles

Frequency Analysis of Monoalphabetic Ciphers

**INSERT
VIDEO
HERE**

- Lesson 1 – Single-character frequencies.
 - Overview
 - Practical hurdles
- Lesson 2 – Multi-character frequencies.

Frequency Analysis of Monoalphabetic Ciphers

**INSERT
VIDEO
HERE**

- Lesson 1 – Single-character frequencies.
 - Overview
 - Practical hurdles
- Lesson 2 – Multi-character frequencies.
 - Digraphs
 - Repeated digraphs
- Lesson 3 – Example
 - Short ciphertext
 - Longer ciphertext

Frequency Analysis of Monoalphabetic Ciphers

**INSERT
VIDEO
HERE**

- Lesson 1 – Single-character frequencies.
 - Overview
 - Practical hurdles
- Lesson 2 – Multi-character frequencies.
 - Digraphs

Frequency Analysis of Monoalphabetic Ciphers

**INSERT
VIDEO
HERE**

- Lesson 1 – Single-character frequencies.
 - Overview
 - Practical hurdles
- Lesson 2 – Multi-character frequencies.
 - Digraphs
 - Repeated digraphs

Frequency Analysis of Monoalphabetic Ciphers

**INSERT
VIDEO
HERE**

- Lesson 1 – Single-character frequencies.
 - Overview
 - Practical hurdles
- Lesson 2 – Multi-character frequencies.
 - Digraphs
 - Repeated digraphs
- Lesson 3 – Example

Frequency Analysis of Monoalphabetic Ciphers

**INSERT
VIDEO
HERE**

- Lesson 1 – Single-character frequencies.
 - Overview
 - Practical hurdles
- Lesson 2 – Multi-character frequencies.
 - Digraphs
 - Repeated digraphs
- Lesson 3 – Example
 - Short ciphertext

Frequency Analysis of Monoalphabetic Ciphers

**INSERT
VIDEO
HERE**

- Lesson 1 – Single-character frequencies.
 - Overview
 - Practical hurdles
- Lesson 2 – Multi-character frequencies.
 - Digraphs
 - Repeated digraphs
- Lesson 3 – Example
 - Short ciphertext
 - Longer ciphertext

It's all about randomness

**INSERT
VIDEO
HERE**

It's all about randomness

**INSERT
VIDEO
HERE**

- Eve intercepts truly random plaintext

It's all about randomness

**INSERT
VIDEO
HERE**

- Eve intercepts truly random plaintext
 - Can't be cryptanalyzed (successfully)

It's all about randomness

**INSERT
VIDEO
HERE**

- Eve intercepts truly random plaintext
 - Can't be cryptanalyzed (successfully)
 - Effect is same as one-time pad

It's all about randomness

**INSERT
VIDEO
HERE**

- Eve intercepts truly random plaintext
 - Can't be cryptanalyzed (successfully)
 - Effect is same as one-time pad
- Non-randomness in plaintext conveys meaning

It's all about randomness

**INSERT
VIDEO
HERE**

- Eve intercepts truly random plaintext
 - Can't be cryptanalyzed (successfully)
 - Effect is same as one-time pad
- Non-randomness in plaintext conveys meaning
- Non-randomness in ciphertext gives hints

It's all about randomness

**INSERT
VIDEO
HERE**

- Eve intercepts truly random plaintext
 - Can't be cryptanalyzed (successfully)
 - Effect is same as one-time pad
- Non-randomness in plaintext conveys meaning
- Non-randomness in ciphertext gives hints
- Monoalphabetic cipher keeps all nonrandomness

It's all about randomness

**INSERT
VIDEO
HERE**

- Eve intercepts truly random plaintext
 - Can't be cryptanalyzed (successfully)
 - Effect is same as one-time pad
- Non-randomness in plaintext conveys meaning
- Non-randomness in ciphertext gives hints
- Monoalphabetic cipher keeps all nonrandomness
- Breakable by matching ciphertext structure to estimated plaintext structure.

etaoin shrdlu

**INSERT
VIDEO
HERE**

E	12.702	D	4.253	P	1.929
T	9.056	L	4.025	B	1.492
A	8.167	C	2.782	V	0.978
O	7.507	U	2.758	K	0.772
I	6.966	M	2.406	J	0.153
N	6.749	W	2.360	X	0.150
S	6.327	F	2.228	Q	0.095
H	6.094	G	2.015	Z	0.074
R	5.987	Y	1.974		

Plaintext or ciphertext?

**INSERT
VIDEO
HERE**

Plaintext or ciphertext?

**INSERT
VIDEO
HERE**

500000:	ILSTU	BBSTO	UTLYS	TOODU	PTOHI
500025:	SSPER	MACET	ISUPP	ERATT	HECAP
500050:	STANH	EADAS	IFTHA	TCAPS	TANWE
500075:	REASI	DEBOA	RDNOR	WASST	UBBTH
500100:	EONLY	BANQU	ETERO	NWHAL	ESFLE
500125:	SHTHA	TNIGH	TMING	LINGT	HEIRM
500150:	UMBLI	NGSWI	THHIS	OWNMA	STICA
500175:	TIONS	THOUS	ANDSO	NTHOU	SANDS
500200:	OFSHA	RKSSW	ARMIN	GROUN	DTHED
500225:	EADLE	VIATH	ANSMA	CKING	LYFEA

Emphasizing alternate words reveals plaintext content

**INSERT
VIDEO
HERE**

Emphasizing alternate words reveals plaintext content

**INSERT
VIDEO
HERE**

500000:	ILSTU	BBSTO	UTLYS	TOODU	PTOHI
500025:	SSPER	MACET	ISUPP	ERATT	HECAP
500050:	STANH	EADAS	IFTHA	TCAPS	TANWE
500075:	REASI	DEBOA	RDNOR	WASST	UBBTH
500100:	EONLY	BANQU	ETERO	NWHAL	ESFLE
500125:	SHTHA	TNIGH	TMING	LINGT	HEIRM
500150:	UMBLI	NGSWI	THHIS	OWNMA	STICA
500175:	TIONS	THOUS	ANDSO	NTHOU	SANDS
500200:	OFSHA	RKSSW	ARMIN	GROUN	DTHED
500225:	EADLE	VIATH	ANSMA	CKING	LYFEA

In review – Patterns are key

**INSERT
VIDEO
HERE**

- Match non-randomness in ciphertext to expected non-randomness in plaintext

In review – Patterns are key

**INSERT
VIDEO
HERE**

- **Match non-randomness in ciphertext to expected non-randomness in plaintext**
- **Statistical variability and intentional efforts to distort patterns frustrate the cryptanalyst**

In review – Patterns are key

**INSERT
VIDEO
HERE**

- **Match non-randomness in ciphertext to expected non-randomness in plaintext**
- **Statistical variability and intentional efforts to distort patterns frustrate the cryptanalyst**
- **Recognizing patterns is an art as well as a skill**

In review – Patterns are key

**INSERT
VIDEO
HERE**

- **Match non-randomness in ciphertext to expected non-randomness in plaintext**
- **Statistical variability and intentional efforts to distort patterns frustrate the cryptanalyst**
- **Recognizing patterns is an art as well as a skill**
- **Next – Multi-character frequency patterns.**